LAB Assignment

(Studio 3T)

ANKITA DAS

CSF 017

COLLECTION 1: **flights**

Schema

{

  flight_id: Number,

  airline: String,

  source: String,

  destination: String,

  price: Number,

  duration: Number

}

Table

flight_id	airline	source	destination	price	duration
1	Indigo	Delhi	Mumbai	5000	2
2	AirIndia	Kolkata	Chennai	6500	3
3	SpiceJet	Mumbai	Bangalore	4000	1.5
4	Vistara	Delhi	Chennai	7000	3
5	GoFirst	Mumbai	Delhi	4800	2
6	Indigo	Bangalore	Kolkata	5500	2.5

COLLECTION 2: **bookings**

Schema

{

  booking_id: Number,

  passenger_name: String,

  flight_id: Number,

  city: String,

  seats: Number,

total_fare: Number

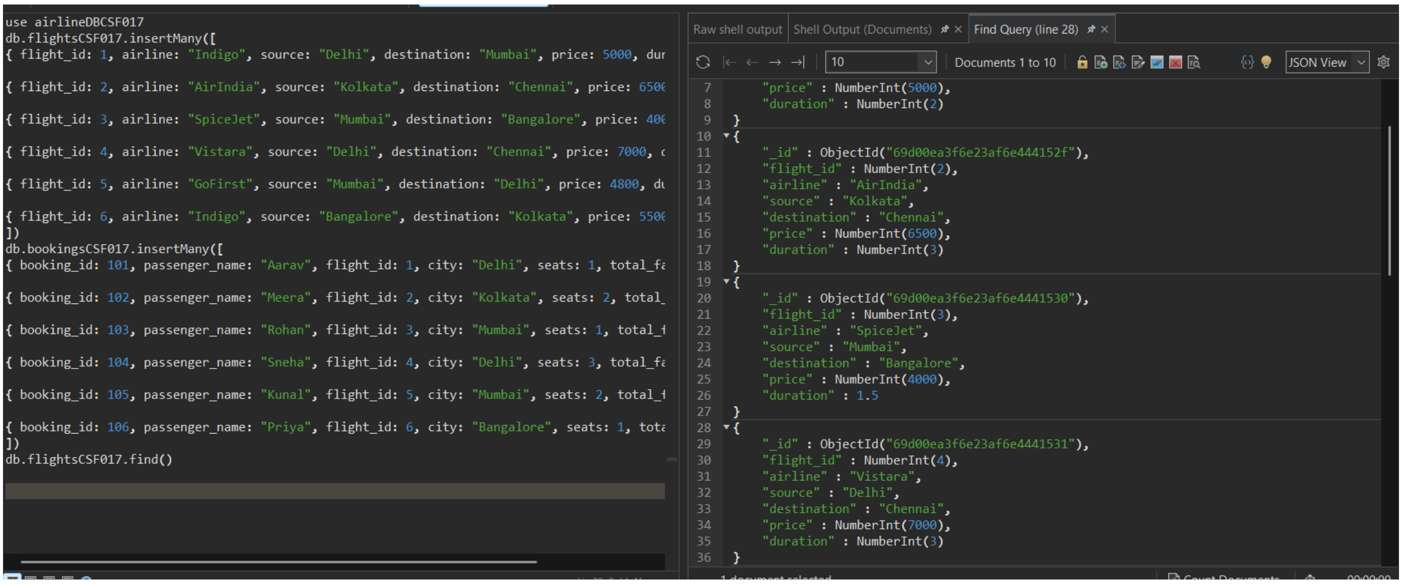
}

Table

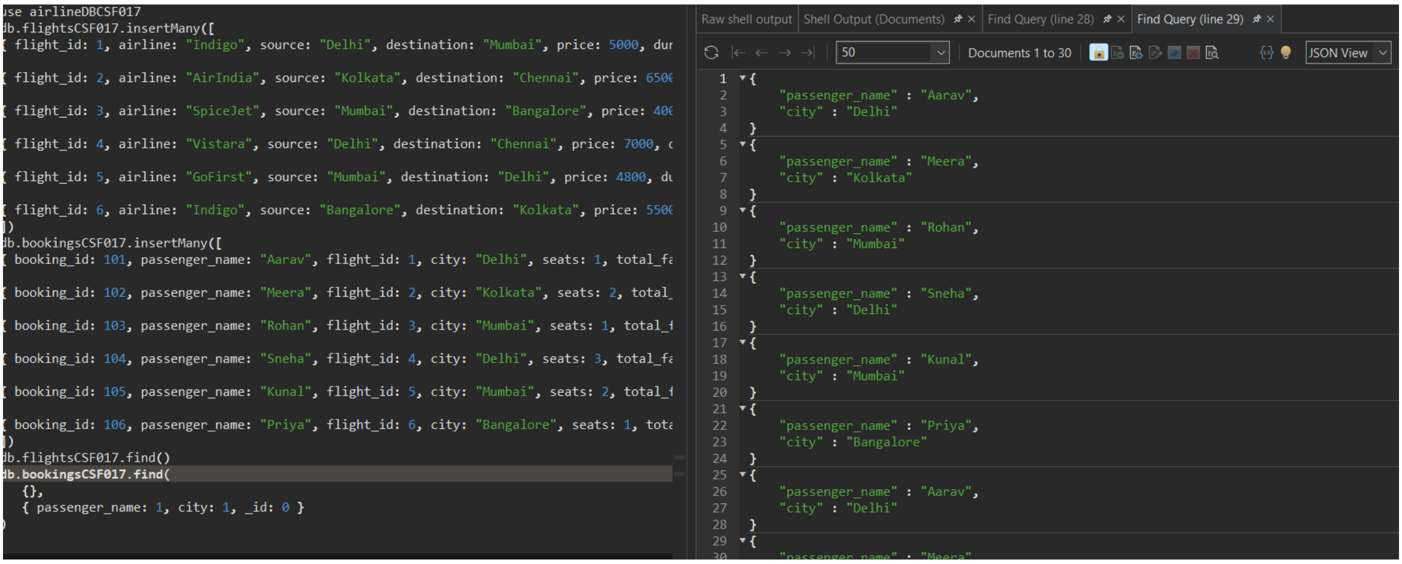
booking_id	passenger_name	flight_id	city	seats	total_fare
101	Aarav	1	Delhi	1	5000
102	Meera	2	Kolkata	2	13000
103	Rohan	3	Mumbai	1	4000
104	Sneha	4	Delhi	3	21000
105	Kunal	5	Mumbai	2	9600
106	Priya	6	Bangalore	1	5500

QUESTIONS

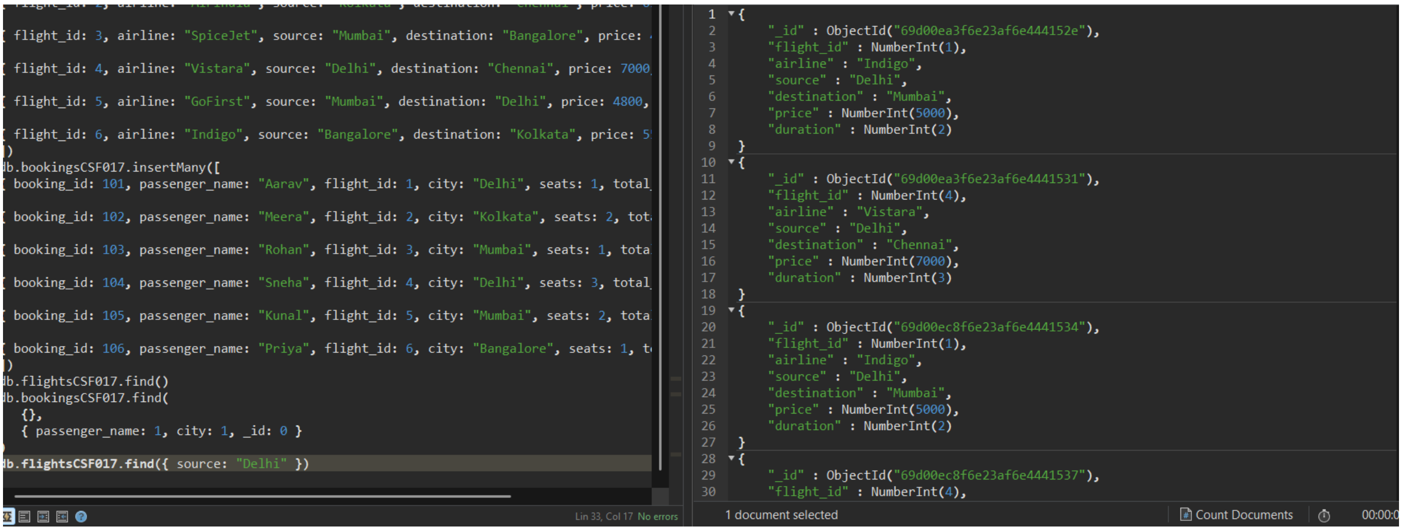
Q1. Display all flights.



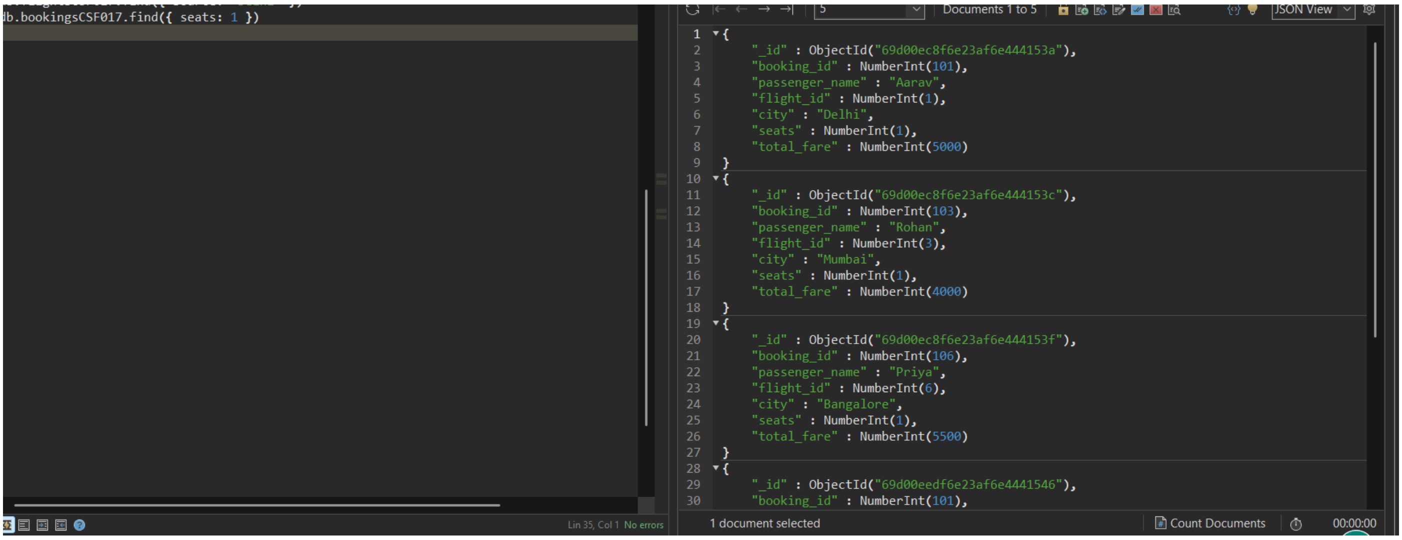
Q2. Display passenger_name and city from bookings.



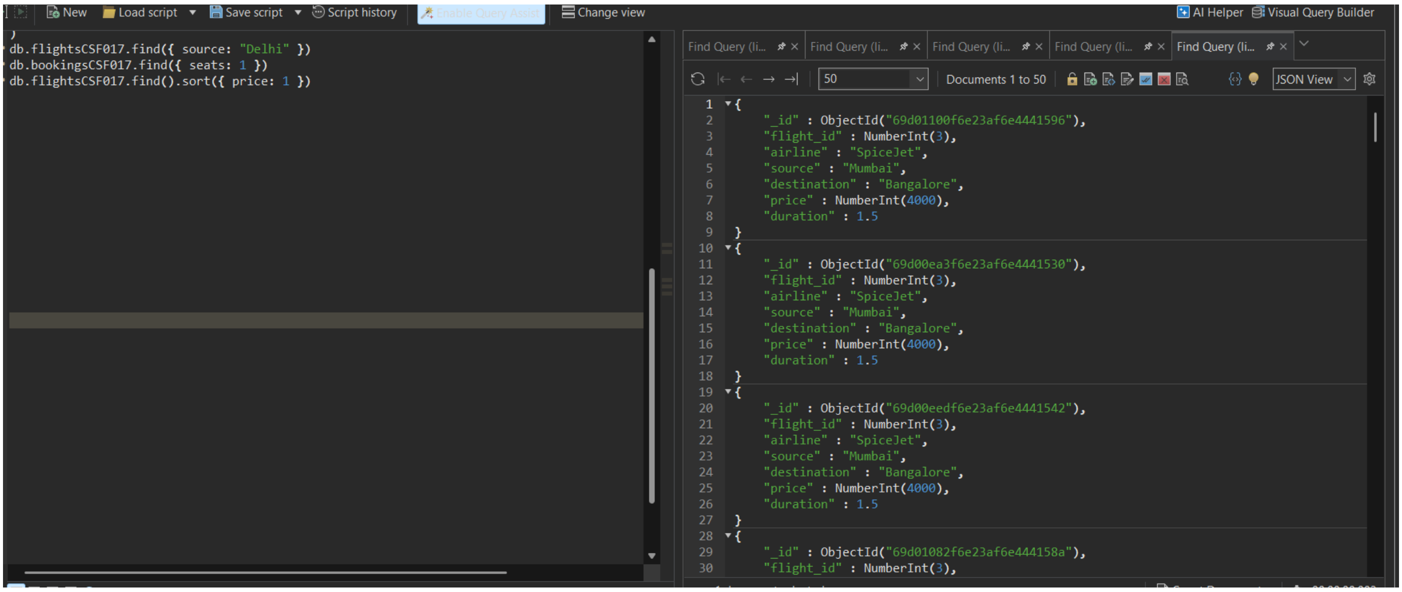
Q3. Find flights where source is Delhi.



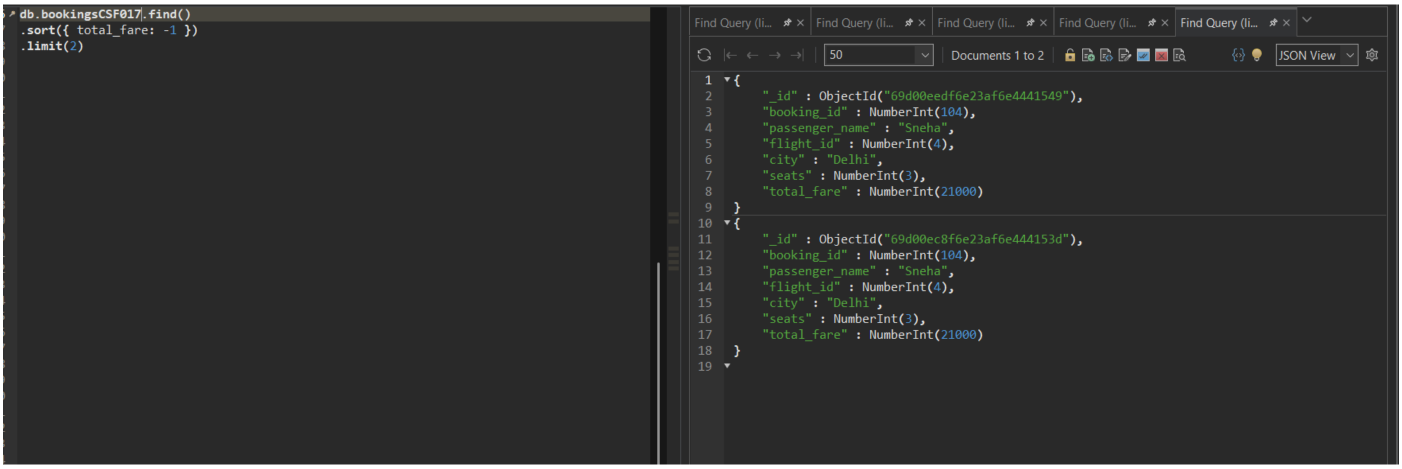
Q4. Find bookings where seats = 1.



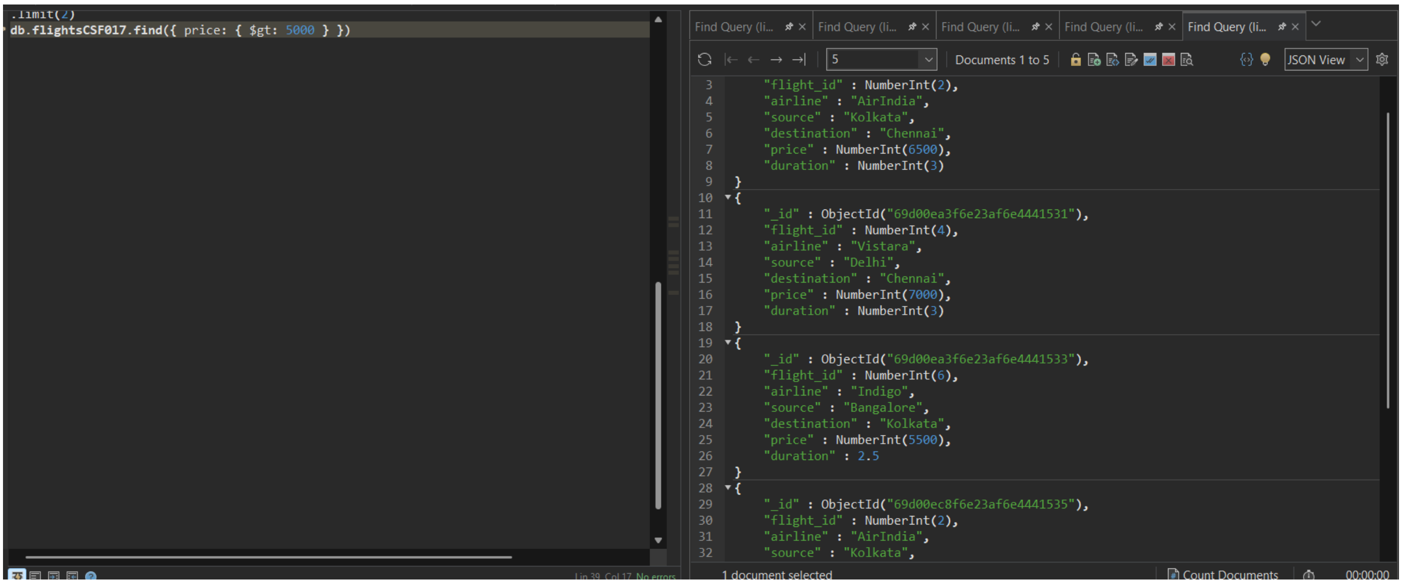
Q5. Sort flights by price in ascending order.



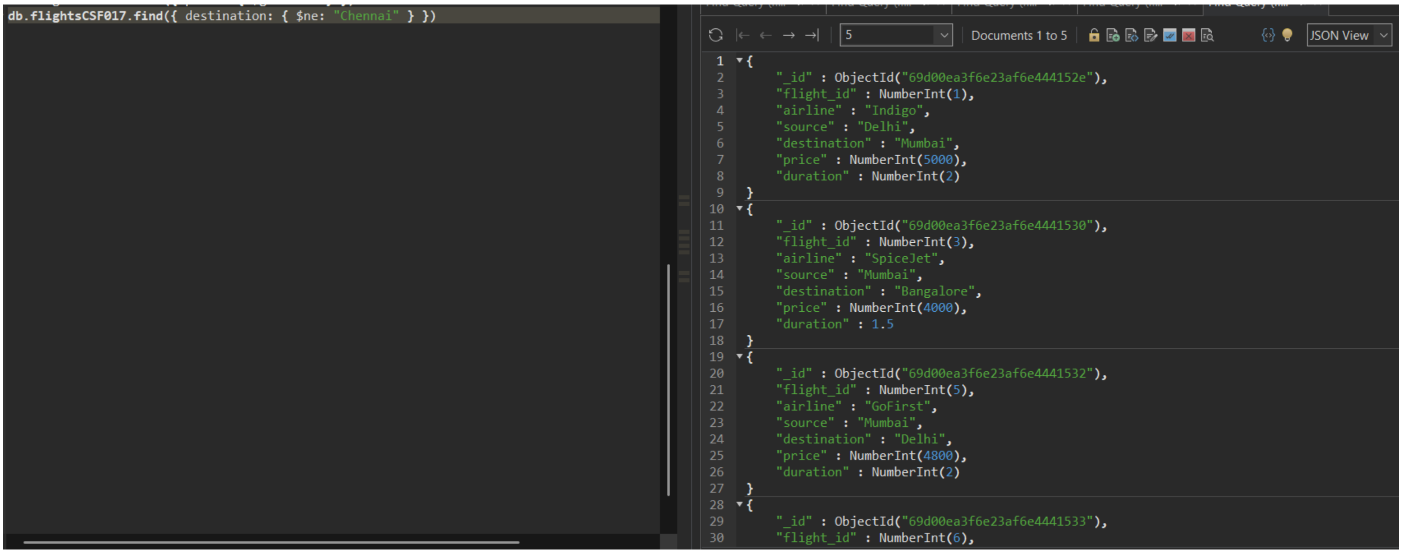
Q6. Display top 2 bookings with highest total_fare.



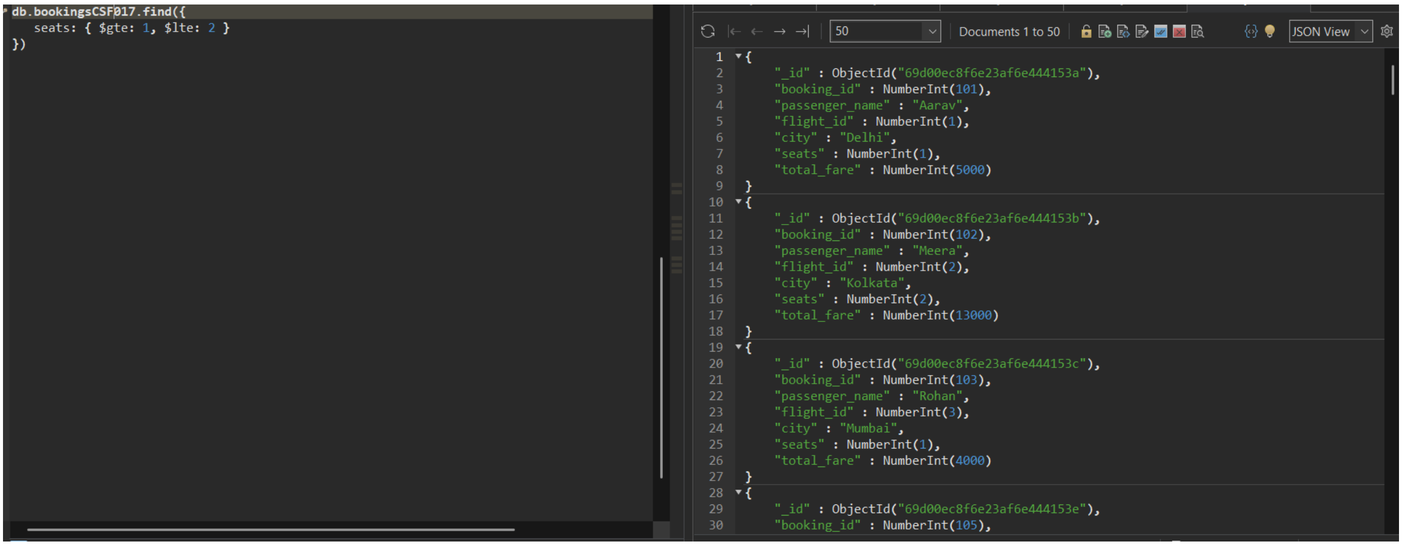
Q7. Find flights where price is greater than 5000.



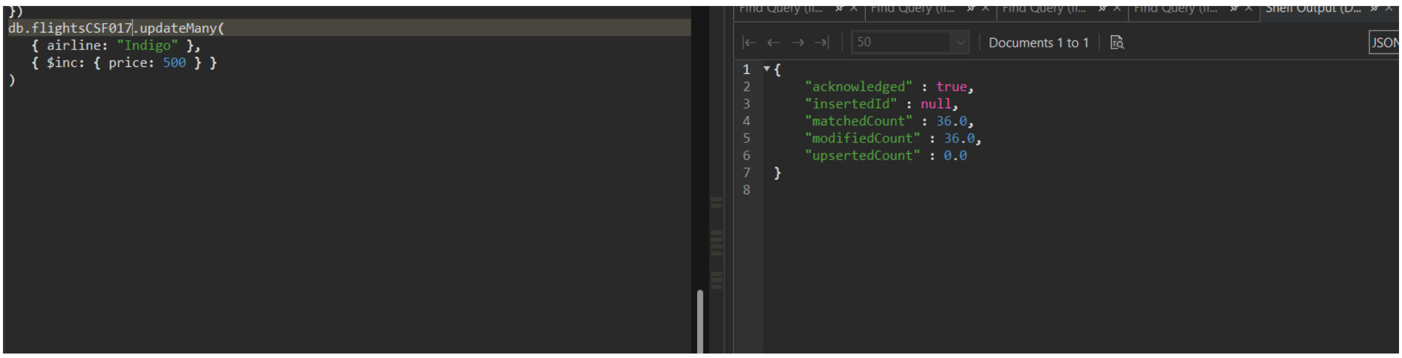
Q8. Display flights where destination is not Chennai.



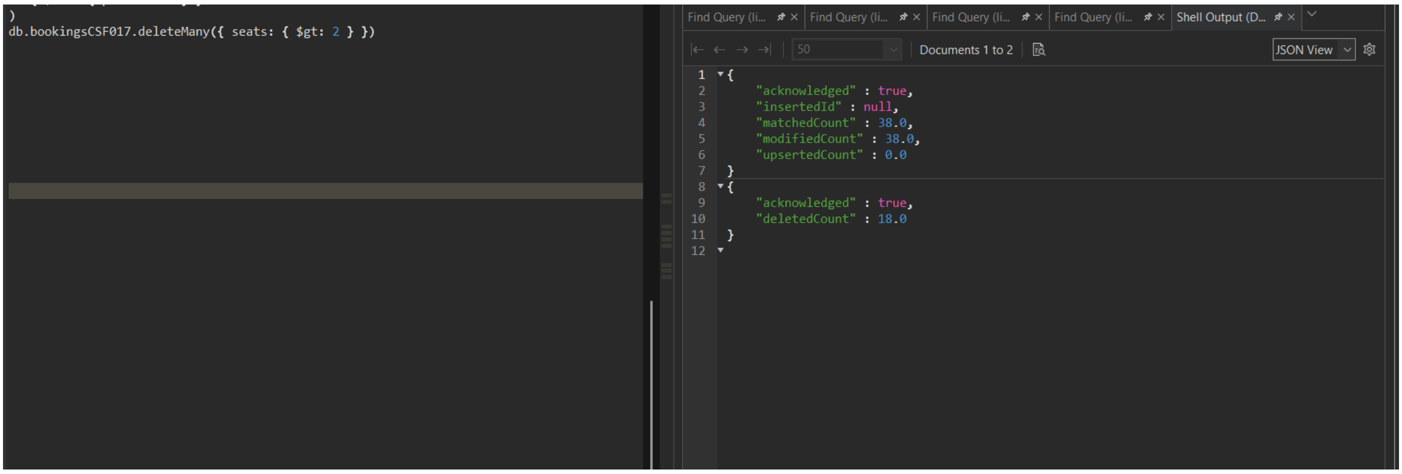
Q9. Find bookings where seats are between 1 and 2.



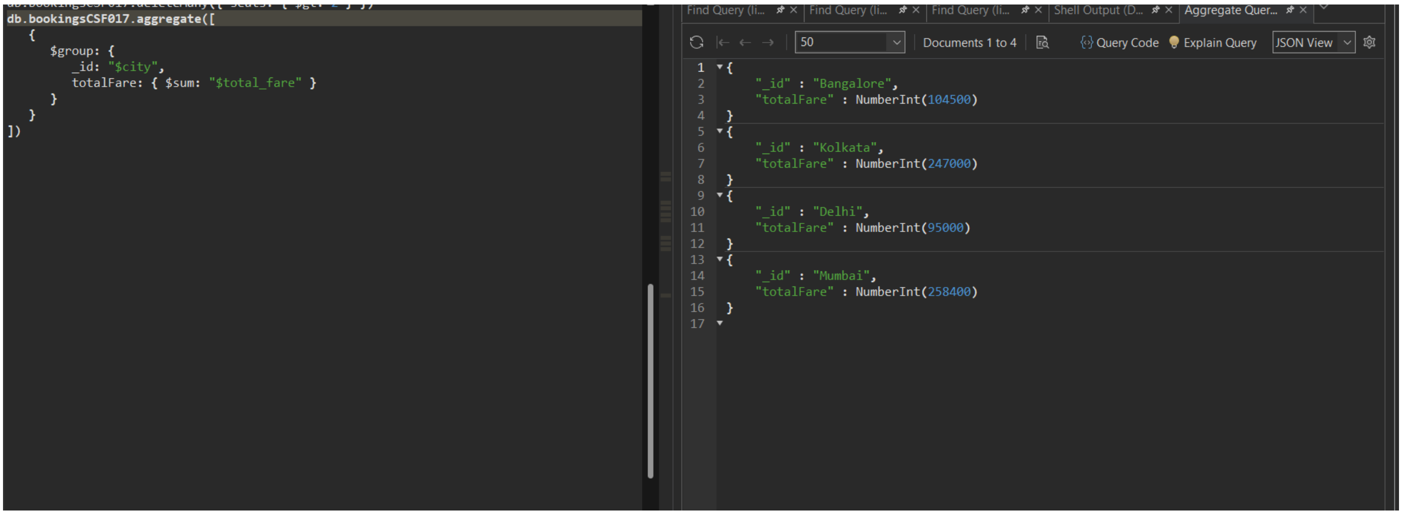
Q10. Update price by +500 for flights of airline "Indigo".



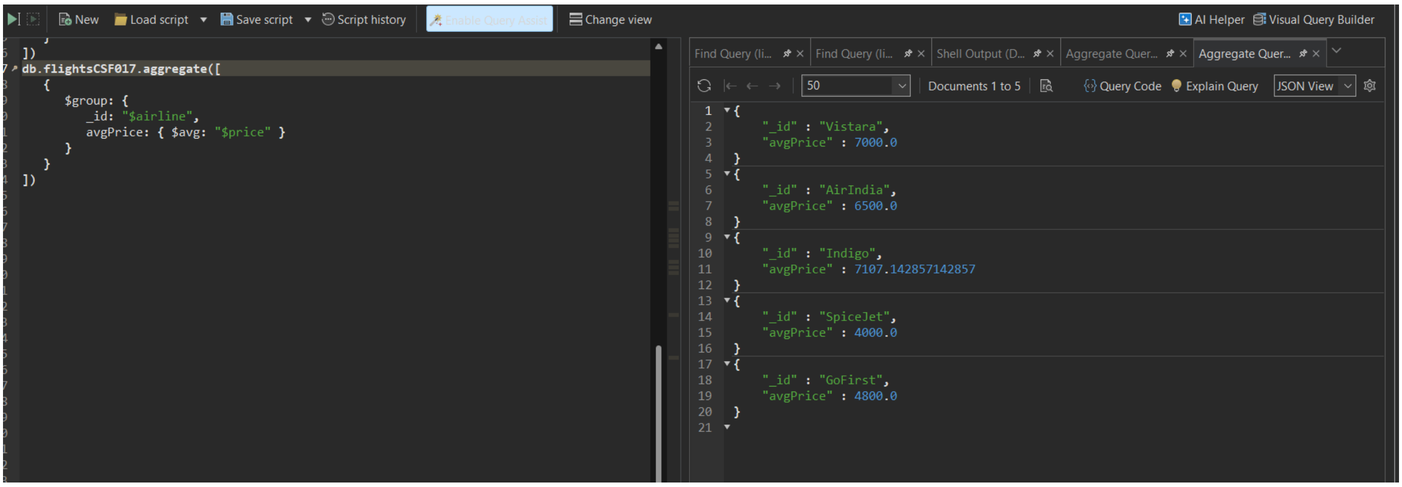
Q11. Delete bookings where seats are greater than 2.



Q12. Find total fare collected city-wise. *(Aggregation)*



Q13. Find average flight price per airline. *(Aggregation)*



Q14. Display airlines where average price is greater than 5500. *(Aggregation + filter)*

```
db.flightsCSF017.aggregate([
    {
        $group: {
            _id: "$airline",
            avgPrice: { $avg: "$price" }
        }
    }
])
d
```

```
Documents 1 to 5
{
    "_id" : "Vistara",
    "avgPrice" : 7000.0
}
{
    "_id" : "AirIndia",
    "avgPrice" : 6500.0
}
{
    "_id" : "Indigo",
    "avgPrice" : 7107.142857142857
}
{
    "_id" : "SpiceJet",
    "avgPrice" : 4000.0
}
{
    "_id" : "GoFirst",
    "avgPrice" : 4800.0
}
```

Q15. Find maximum total_fare among bookings for each city. *(Aggregation)*

```
db.bookingsCSF017.aggregate([
    {
        $group: {
            _id: "$city",
            maxFare: { $max: "$total_fare" }
        }
    }
])
```

```
Documents 1 to 4
{
    "_id" : "Kolkata",
    "maxFare" : NumberInt(13000)
}
{
    "_id" : "Delhi",
    "maxFare" : NumberInt(5000)
}
{
    "_id" : "Mumbai",
    "maxFare" : NumberInt(9600)
}
{
    "_id" : "Bangalore",
    "maxFare" : NumberInt(5500)
}
```